

INDUSTRIAL BATTERY TEST ANALYTICS

Engineering Report

BMS Monitoring & Performance Analysis

Test ID	INEVYD51105B0012
Test Profile	100% DOD B-SMART
Dataset	INEVYD51105B0012 100% DOD B-SMART 06.05.2025.xlsx
Test Start	2025-05-06 15:52:32
Test End	2025-05-07 01:27:42
Duration	9.59 h
Total Records	34,512
Cycles	2
Cell Count	16
Report Date	2026-06-18 23:54

1. Data Quality Summary

Overall: PASS ✓

Layer	Rows	Cols	Dups	Miss Cols
CYCLE	2	22	0	1
STEP	11	31	0	14
RECORD	34512	99	0	2
EVENT	13	4	0	0

2. Battery Performance Summary

Charge Capacity	110.503 Ah
Discharge Capacity	110.185 Ah
Charge Energy	5967.0 Wh
Discharge Energy	5669.7 Wh
Coulombic Efficiency	99.712 %
Energy Efficiency	95.016 %
Charge Mid-Voltage	55.663 V
Discharge Mid-Voltage	46.125 V
Charge Duration	4.31 h
Discharge Duration	3.77 h
Peak Charge Current	30.0 A (0.27C)
Peak Discharge Current	30.0 A (0.27C)
Avg Charge Power	1388.9 W
Avg Discharge Power	1502.9 W
Peak Pack Voltage	61.116 V
Min Pack Voltage	11.079 V

Coulombic Efficiency above 99.5% indicates minimal parasitic side reactions. Energy Efficiency reflects total round-trip losses including ohmic heating. C-rate is referenced to the measured discharge capacity.

3. Charge–Discharge Profile Analysis

3.1 Voltage & Current Profile

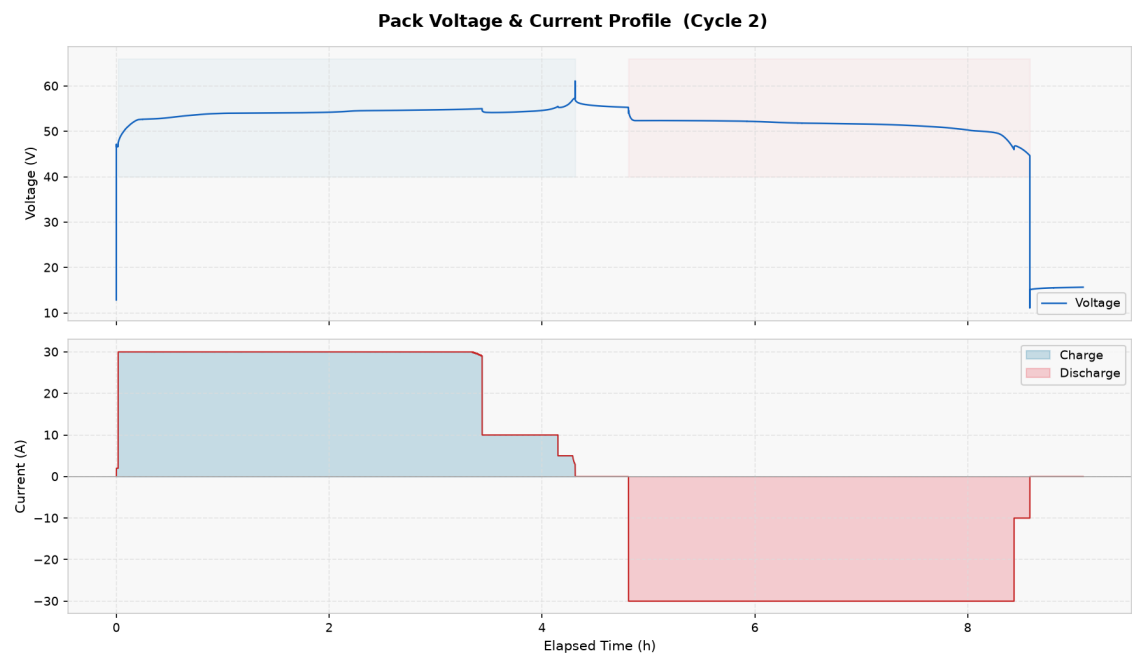


Figure 1 — Pack voltage and current vs elapsed time.

3.2 Power Profile

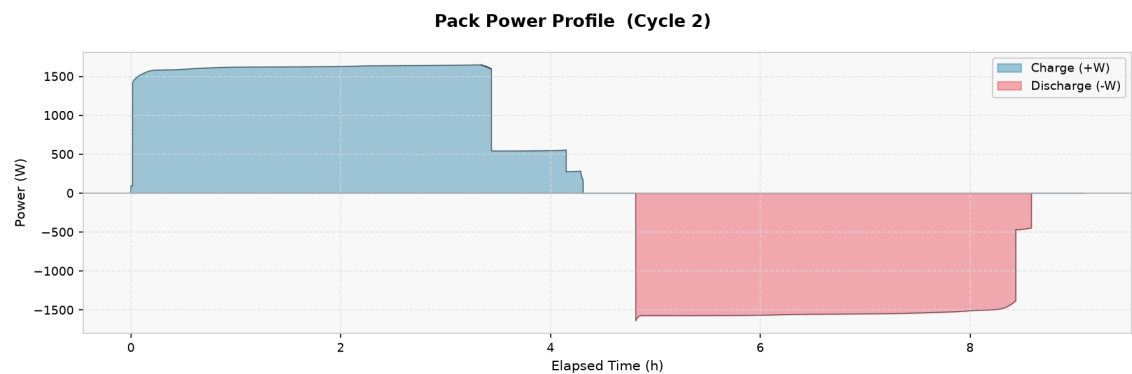


Figure 2 — Pack power profile. Positive = charge; Negative = discharge.

3.3 Cumulative Capacity & Energy

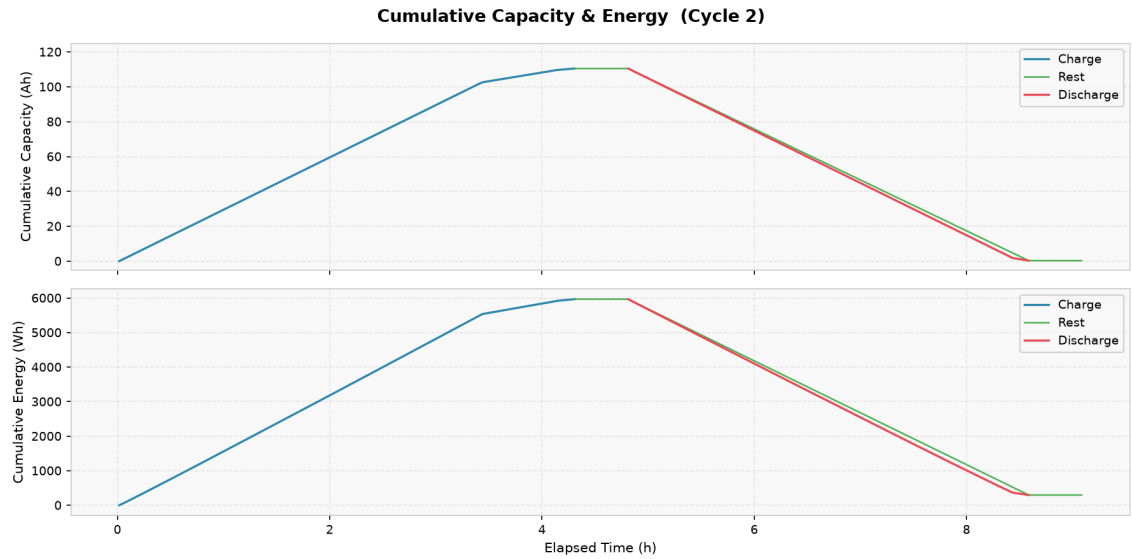


Figure 3 — Cumulative capacity (0 to ~110 Ah) and energy (0 to ~5967 Wh).

3.4 Summary Metrics

Charge start voltage	47.299 V
Charge end voltage	61.116 V
Charge peak power	1647.9 W
CC phase duration	4.13 h
CV phase duration	0.16 h
Charge mid-voltage	55.663 V
Discharge start voltage	55.078 V
Discharge end voltage	11.079 V
Discharge peak power	1637.3 W
Discharge avg power	1502.9 W
Discharge mid-voltage	46.125 V
Rest post-charge V	55.659 V
Rest post-discharge V	15.510 V
Voltage hysteresis	9.538 V

4. Cell Voltage Analysis

4.1 Individual Cell Voltages

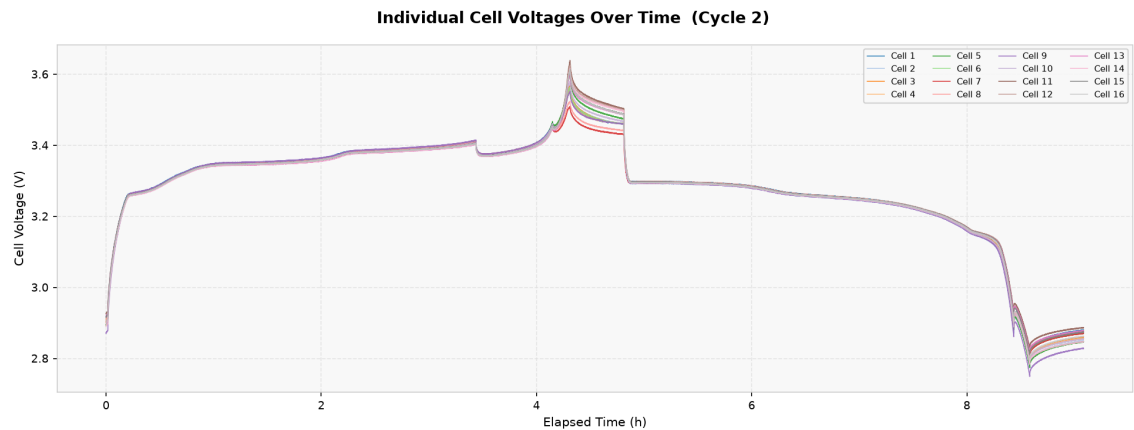


Figure 4 — All 16 cell voltages over the full test duration.

4.2 Voltage Spread

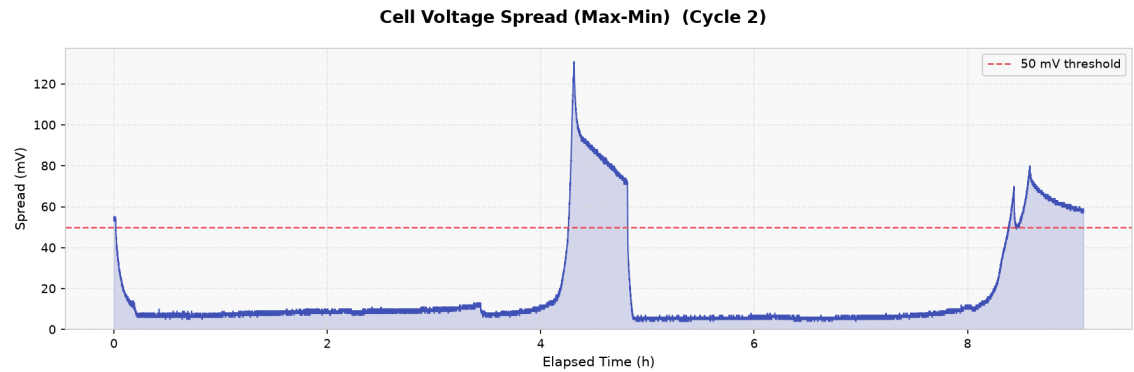


Figure 5 — Cell voltage spread with 50 mV imbalance threshold.

4.3 Per-Cell Average Voltage

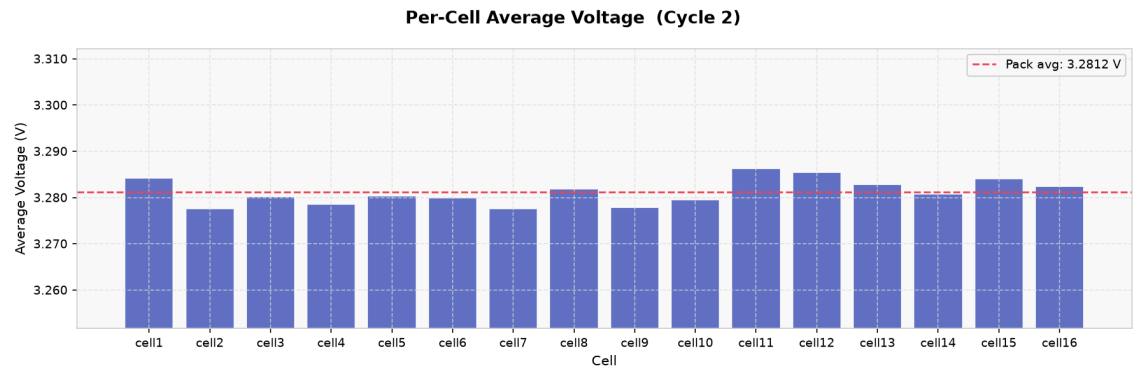


Figure 6 — Per-cell average voltage. Red dashed line = pack average.

4.4 Imbalance Summary

Average spread	17.6 mV
Peak spread	131.0 mV
Minimum spread	4.0 mV
Weakest cell	cell2_V (avg 3.2776 V)

Strongest cell	cell11_V (avg 3.2863 V)
Cell-to-cell avg delta	8.6 mV
Imbalance flag (>50mV)	NO

5. Thermal Analysis

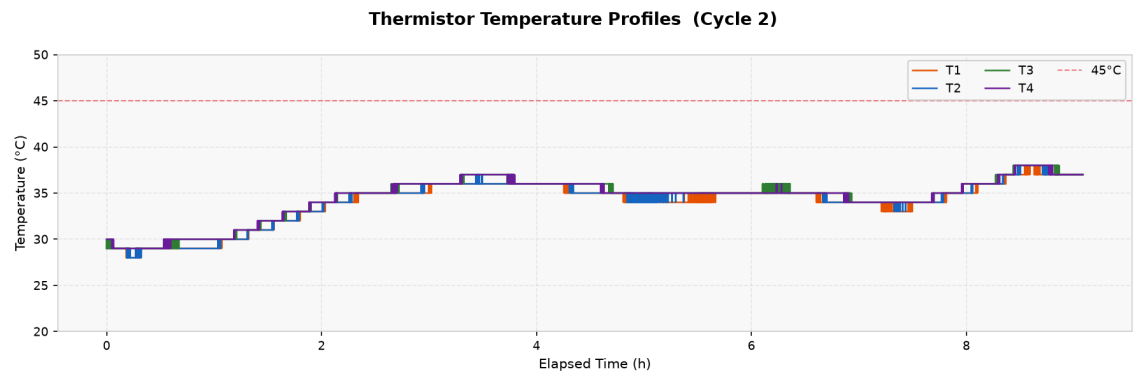


Figure 7 — Thermistor temperature profiles over test. 45 C threshold shown.

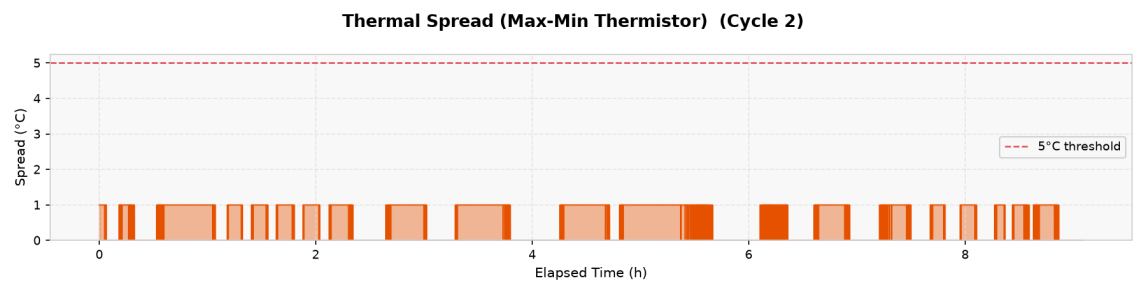


Figure 8 — Sensor-to-sensor thermal spread. 5 C threshold shown.

Peak temperature	38.0 C
Minimum temperature	28.0 C
Average temperature	34.3 C
Std deviation	2.37 C
Avg sensor spread	0.52 C
Max sensor spread	1.00 C
Avg temp during charge	33.2 C
Avg temp during discharge	34.9 C
Avg temp during rest	36.3 C
BMS max temp (avg)	34.5 C
Hotspot flag (>=45 C)	NO
Spread flag (>5 C)	NO

6. BMS Telemetry Analysis

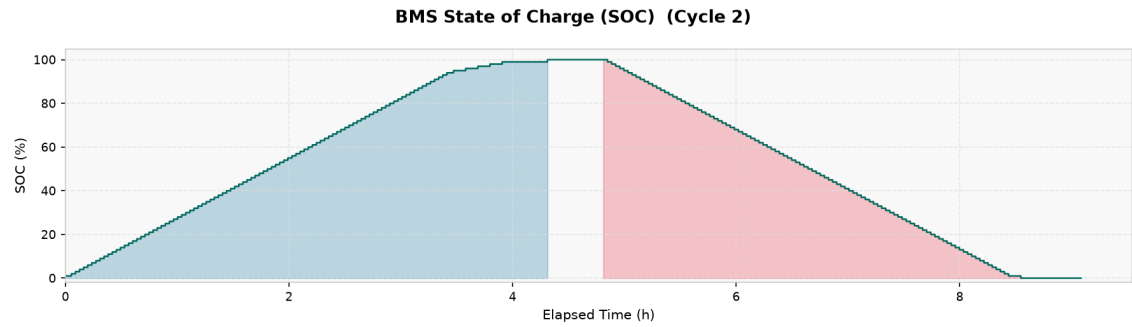


Figure 9 — BMS SOC: rises 0 to 100% during charge; falls 100 to 0% during discharge.

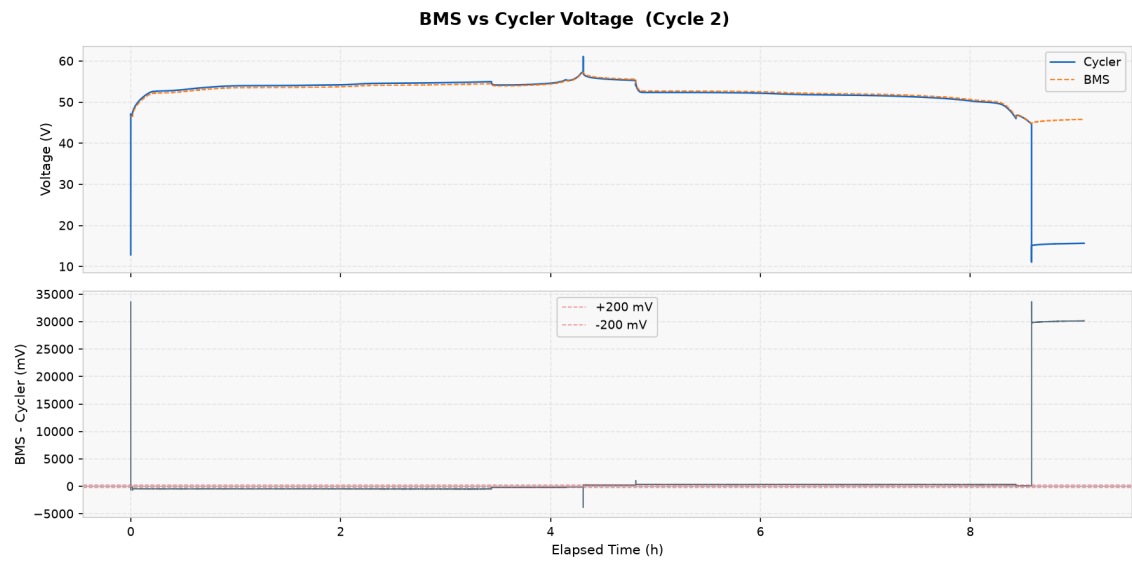


Figure 10 — BMS vs cycler voltage overlay and discrepancy trace (mV).

SOC start	0.0 %
SOC end	0.0 %
SOC range	0.0 – 100.0 %
Residual cap (peak)	110.035 Ah
Residual cap (start)	0.003 Ah
Residual cap (end)	0.000 Ah
BMS avg voltage	52.503 V
Cycler avg voltage	50.892 V
Voltage discrepancy	+1610.6 mV
BMS cycle count	6 to 7

Residual_Capacity(AH) is the BMS remaining-capacity register (0 to 110 Ah). Voltage discrepancy reflects sensing-point differences between BMS terminals and cycler cable measurement points. Values within +/-200 mV are acceptable.

7. Protection & Safety Analysis

Protection columns in this B-SMART export are binary status flags (0 = not triggered, 1 = triggered). No threshold values are exported by this cycler software. Operating margins are estimated against typical LFP BMS reference values.

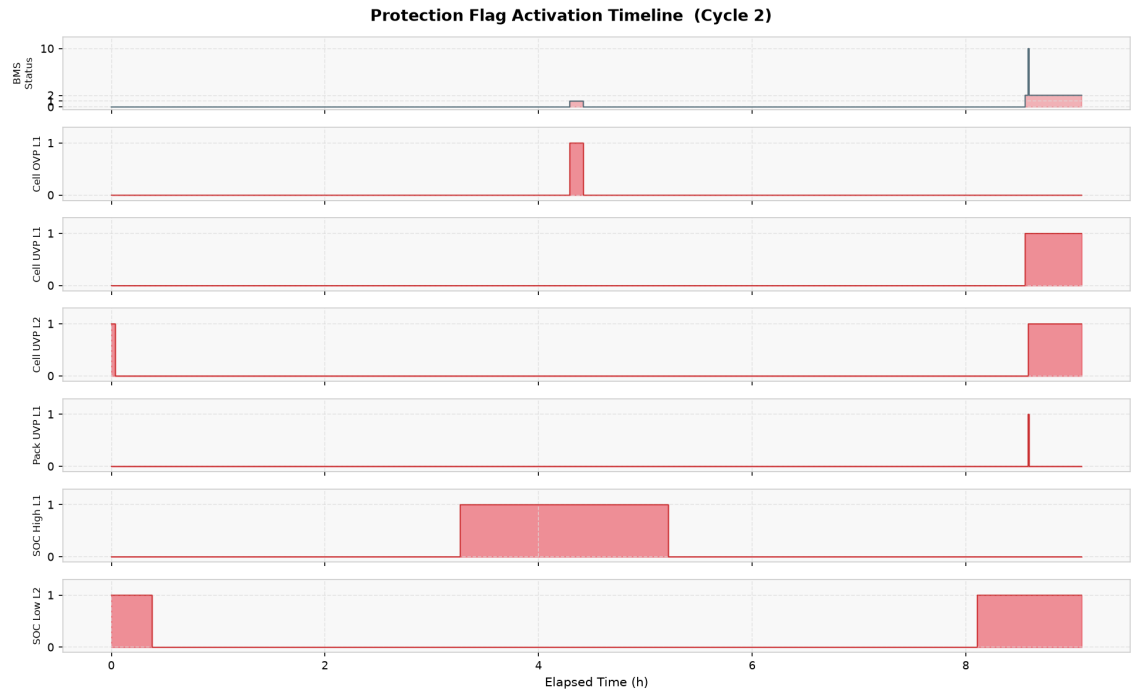


Figure 11 — Protection flag activation timeline and BMS_Status over Cycle 2.

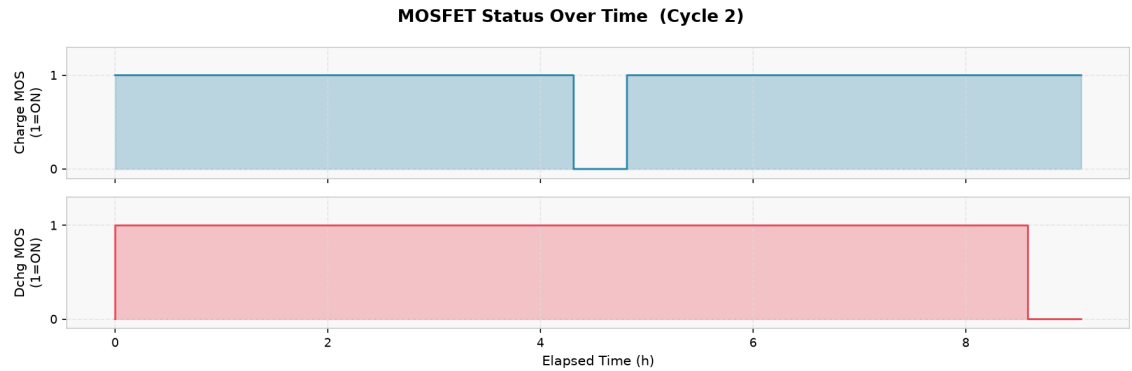


Figure 12 — Charge and discharge MOSFET states over Cycle 2.

Peak cell voltage	3.639 V
Min cell voltage	2.749 V
Peak temperature	38.0 C
Peak charge current	30.0 A
Peak discharge current	30.0 A
Est. OVP headroom	11 mV (ref 3.65 V/cell)
Est. UVP headroom	249 mV (ref 2.50 V/cell)
Est. OTP headroom	7.0 C (ref 45 C)
Triggered flags	Cell OVP L1, Cell UVP L1, Cell UVP L2, Pack UVP L1, SOC High L1, SOC Low L2

Safety verdict	FAIL
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- BMS OVP events: 458 records.

8. Key Findings

Performance

- Coulombic Efficiency 99.712% — Excellent ($\geq 99\%$)
- Energy Efficiency 95.016% — Good ($\geq 90\%$)
- Battery delivered 110.185 Ah at 0.27C with discharge mid-point voltage 46.125 V.

Cell Uniformity

- Average spread 17.6 mV — Within limits (< 50 mV)
- Weakest cell: cell2_V at 3.2776 V avg. Strongest: cell11_V at 3.2863 V avg.

Thermal Behaviour

- Peak temperature 38.0 C — Within normal range (< 45 C)
- Max sensor spread 1.00 C — Uniform (< 5 C spread)

Safety

- Overall safety verdict: FAIL
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9. Recommendations & Limitations

Recommendations

- Monitor the weakest cell over subsequent cycles to detect early capacity divergence.
- Verify BMS passive balancing activation during the CV phase if cell spread increases.
- Schedule a DCIR measurement after completing additional cycles to establish a resistance baseline.
- Cross-validate BMS voltage calibration against the cycler periodically.
- Validate thermal management adequacy for ambient conditions above 30 C.

Limitations

- Single test cycle (100% DOD B-SMART). Cycle-life trends require multi-cycle data.
- BMS telemetry sample rate is lower than cycler record rate; fast transients may not be captured.
- No Reference Performance Test (RPT) data available; all values are interpreted in isolation.
- Temperature coverage is limited to four thermistors; full spatial distribution is not characterised.
- Protection threshold values are not exported by this cycler; margins are engineering estimates only.
- SOH, RUL, and degradation modelling are outside the scope of this report.